

Response to TCE-2026-04-1669

AURORA: A Multi-Objective Reinforcement Learning Framework for Adaptive Cognitive State Optimization in Virtual Reality

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and Khaza Anuarul Hoque, *Senior Member, IEEE*

This report is in response to the comments obtained from the reviewers. According to the comments from the reviewers, we have revised the manuscript and have addressed all of the issues that the reviewers raised to the best of our ability (highlighted in blue in the revised manuscript). We would like to thank the reviewers for their constructive comments, which helped us significantly improve the quality of the paper.

I. RESPONSES TO THE COMMENTS FROM REVIEWER-1

Concern 1: Experimental order effects and lack of counterbalancing. The most important concern is the experimental design, particularly the apparent fixed order of the main experimental conditions. According to the current description, the experiment was conducted sequentially as baseline first, static visual techniques next, AURORA thereafter, and AURORA-HIL finally on a separate day. Although the order of the four static techniques appears to have been randomized within the static condition, the four main conditions do not appear to have been randomized or counterbalanced across participants. If the order was fixed, the observed improvements in AURORA and AURORA-HIL may not be solely attributable to the proposed framework. This concern is particularly important for AURORA-HIL because it was evaluated with the same participants on a separate day after prior exposure to the VR task and after data from earlier conditions had been used for personalization.

Response: We thank the reviewer for this comment and apologize for the confusion. In our experimental conditions, we applied counterbalancing to reduce order effects: the first three

conditions (e.g., Baseline, Static visual techniques, and AURORA) were counterbalanced across participants. However, the static visual technique subconditions (e.g., DFOV, DGB, DOF, and RF) within Condition 2 were presented in randomized order to minimize potential ordering effects among these four static visual techniques, following prior works [1, 2]. The only exception was Condition 4, AURORA-HIL, which was necessarily conducted last, on a separate day, because human-in-the-loop RL requires prior interaction data from which to learn user-specific preferences before the policy can be adapted. Since the fine-tuning is performed offline via PPO updates on data collected during the AURORA condition, the personalized policy first had to be trained before it could be evaluated in the following session (AURORA-HIL).

It is worth mentioning that prior work has shown that cybersickness symptoms can persist from hours up to several days after VR exposure (<https://doi.org/10.1007/s10055-016-0285-9>, <https://doi.org/10.3357/ASEM.2394.2009>), suggesting that participants may still experience residual cybersickness symptoms the following day. Consistent with this, in our study, a few participants felt discomfort both on the day of Conditions 1–3 and continued to experience it on the following day, yet reported substantially greater comfort during the AURORA-HIL condition. For instance, after AURORA-HIL, one participant noted: *“I was still feeling uncomfortable from the previous day, but during today’s session I felt more comfortable, and after the session I felt better than I had the day before.”*. It is important to note that the sequential wording in the paper (e.g., Initially, Next, Then, and Finally) was intended only to describe the order in which conditions are introduced in the manuscript, not the actual exposure order in the user study. We have revised Section VI to remove the sequential phrasing, explicitly clarify the scope of counterbalancing, explain the rationale for evaluating AURORA-HIL separately after the non-HIL conditions, and added participant quotes to Section VI(B) in our revised manuscript.

Concern 2: Insufficient adaptive and ablation baselines. Another major concern is that the comparison baselines are not strong enough to support the authors’ broad claims about the necessity and superiority of multi-objective RL. The manuscript compares AURORA mainly against baseline and static visual techniques. Although the comparison between AURORA and AURORA-HIL provides a preliminary assessment of personalization, the manuscript still lacks systematic ablations and stronger adaptive baselines, such as a rule-based adaptive system, single-objective RL for cybersickness only, MTL prediction without RL adaptation, fixed-intensity adaptive policies, and random or greedy policy

baselines. Without these comparisons, it remains unclear whether the observed benefits come specifically from the proposed multi-objective RL formulation, from simple adaptivity, from continuous intensity adjustment, from HIL personalization, or from repeated task exposure.

Response: We thank the reviewer for this comment. In the following we present our response to this.

Regarding the rule-based adaptive system and fixed-intensity adaptive policy: Prior cybersickness mitigation approaches can be broadly categorized as static or rule-based adaptive methods. In static approaches, a single mitigation technique is applied at a fixed, predefined intensity regardless of the user’s current cognitive state [3–8]. In contrast, rule-based adaptive methods adjust mitigation based on a specific predicted signal, such as cybersickness severity or optical flow, using predefined thresholds or handcrafted mappings [1, 2, 9–11]. For instance, explainable AI (XAI)-guided rule-based methods [1, 2] use model explanations to identify important features and trigger predefined mitigation actions. However, these methods remain limited because the mitigation policy is manually designed: the selected technique, activation threshold, and intensity are fixed in advance and cannot adapt continuously to the user’s evolving cognitive and physiological state. Moreover, existing static and rule-based approaches primarily target cybersickness alone and do not jointly account for other important cognitive states, such as cognitive load, attention, and working memory [1–10]. As a result, they cannot balance multiple competing objectives, such as reducing cybersickness while balancing cognitive load and enhancing working memory capabilities. In some cases, an aggressive mitigation response may reduce cybersickness symptoms but increase cognitive load or interfere with working memory, thereby degrading the overall VR experience. Therefore, we do not consider these rule-based dynamic methods as fully adaptive in the sense of our work, because they lack a learned policy that continuously balances user other cognitive states during immersion, as clarified in the abstract and Section I. Accordingly, to evaluate our proposed framework, we compare AURORA against static visual techniques in Section VI as the fixed-intensity baseline, where each technique is applied at a predefined intensity without regard to the user’s real-time predicted cognitive state. While a direct comparison against dynamic mitigation approaches is not included in the current user study, we acknowledge this as a limitation and identify it as a direction for future work in Section VII.

Regarding single-objective RL for a cybersickness-only baseline: A single-objective RL agent

optimizes a single cognitive state (e.g., cybersickness) while ignoring the competing demands of other cognitive states (e.g., cognitive load and working memory). For instance, applying a narrowing field-of-view restriction can reduce cybersickness but may simultaneously increase cognitive load and impair working memory [12–14]. As a result, a single-objective RL agent may improve its targeted cognitive state while negatively affecting the remaining cognitive states. This is precisely why AURORA is designed to jointly optimize all four cognitive states through a multi-objective RL policy guided by an MTL-enabled DL-based prediction model for reward shaping, as detailed in Sections I and III. Furthermore, a single-objective RL design is less efficient because optimizing each cognitive state separately requires both a dedicated prediction model and a dedicated RL agent for each cognitive state. Whereas our MTL-enabled DL-based prediction model uses a single shared backbone with four lightweight output heads to predict all four cognitive states at once, a single-task alternative would need a separate full backbone for each cognitive state, and each corresponding RL agent would have to be trained independently on its own objective. Consequently, both the training cost and the deployed memory and computation requirements would grow by approximately $4\times$, which is a significant concern for deployment on resource-constrained consumer-grade XR headsets. In addition, the four agents would each recommend a separate action for their own target state, which can create conflicts in visual technique and intensity selection. For example, the single-objective RL agent trained for cybersickness reduction might select a high DFOV intensity to suppress cybersickness, while the single-objective RL agent trained for working memory improvement selects a lower DFOV intensity to preserve working memory, leaving no consistent action to apply without an additional arbitration step at each decision point. Therefore, to compare single-objective RL with our proposed multi-objective RL agent, and following the reviewer’s suggestion, we train four single-objective agents, each targeting only one cognitive state, namely cybersickness, cognitive physical load, cognitive mental load, or working memory, under the same architecture, action space, and environment as AURORA, with the only difference being the training reward, which is restricted to the single targeted state. All RL agents are then evaluated using the full multi-objective reward in Equation 2, as this defines the complete optimization objective of AURORA. Each single-objective RL agent improves its own targeted cognitive state but achieves a lower total reward than AURORA, since gains in one cognitive state come at the expense of the others, while AURORA balances all four cognitive states jointly and achieves the highest total reward, confirming that the multi-objective formulation, rather than single-state

optimization, is responsible for the observed improvements. We have added the single-objective agent comparison in Section IV and the updated Fig. 2, together with the inference model sizes. Each single-task DL-based prediction model is approximately 13.2 MB (cybersickness), 12.8 MB (cognitive physical load), 13.0 MB (cognitive mental load), and 12.6 MB (working memory), totaling about 51.6 MB, and each single-objective RL agent is approximately 2.4 MB in ONNX format, totaling about 9.6 MB across the four agents. In comparison, our single MTL-enabled DL-based prediction model is approximately 14.5 MB, and the joint multi-objective RL agent is approximately 2.9 MB, confirming the approximately $4\times$ overhead of the single-objective design in both the prediction model and the RL agent.

Regarding MTL prediction without RL adaptation: Although our MTL-enabled DL-based prediction model accurately predicts individual cognitive states (e.g., cybersickness, cognitive physical load, cognitive mental load, and working memory), it does not determine which visual technique to apply or at what intensity. Acting on these predictions without RL would therefore require a fixed, rule-based mapping from predicted cognitive states to visual technique actions. For example, the mapping might apply DFOV when predicted cybersickness is high, switch to DGB when both cybersickness and cognitive physical load are high, or reduce the intensity when cognitive mental load is high or working memory is low, each following a predefined rule at a fixed intensity. However, this kind of mapping is static and predefined, applying a single visual technique through a one-size-fits-all rule that cannot suit every individual, since predicted cognitive states and their responses to applied visual techniques vary considerably across users [13]. Moreover, a fixed mapping responds only to the current prediction and cannot account for the delayed effects of an applied visual technique, where an action that improves one cognitive state now may worsen other cognitive states a few steps later. For example, applying a narrower DFOV may reduce cybersickness immediately, but the added head movement it requires can increase cognitive load a few steps later. An RL agent is therefore needed because it learns from interaction over time, optimizing the long-horizon cumulative outcome rather than only the immediate outcome, so that it can anticipate these delayed effects and continuously adapt both the visual technique and its intensity to each user's predicted cognitive states during immersion. For these reasons, rather than relying on the MTL-enabled DL-based prediction model alone, our proposed framework couples it with an RL agent, where the prediction model provides the cognitive states during immersion and the RL agent learns how to act on them, jointly enabling

adaptive and personalized visual technique selection that prediction alone cannot achieve.

Regarding random and greedy policy baselines: In our work, we use Proximal Policy Optimization (PPO) [15] as our RL algorithm, which is a widely used policy-gradient method for continuous control and multi-objective reinforcement learning [16]. It also aligns well with our setting, as our action space combines a discrete choice of visual technique (DFOV, DGB, DOF, or RF) with a continuous intensity value, and PPO supports both discrete and continuous actions simultaneously while learning a control policy that optimizes the long-horizon cumulative reward over repeated interaction, which allows it to balance the four competing cognitive states and account for the delayed effects of an applied visual technique. In contrast, a random policy selects visual techniques and intensities regardless of the cognitive states or reward, producing a reward that fluctuates near zero and never improves. Similarly, a greedy policy, although more informed than a random one, selects the technique-intensity pair with the highest immediate predicted reward at the current step but sees only that step and cannot account for delayed effects, where an action that reduces one cognitive state now may worsen other cognitive states later. Since neither baseline learns over repeated interaction nor optimizes over a long horizon, neither can jointly balance the discrete visual technique and continuous intensity across the four competing cognitive states, and thus neither supports our multi-objective formulation. To verify this, we evaluate the greedy policy (the stronger of the two baselines) in our domain-randomized simulation using the same full-reward formulation. Unlike PPO, which directly handles continuous intensity, the greedy policy can only choose among a finite set of actions, so we discretize the continuous intensity into a fine grid to allow the greedy policy to select an action at each step. As the greedy policy does not update its parameters, we record its reward at each evaluation step rather than over a training curve. The greedy policy reaches a much lower reward than AURORA and shows no improvement over steps, confirming that the multi-objective improvements of AURORA come from long-horizon RL optimization rather than simple adaptivity. We have added the greedy-policy comparison and its reward curve to the revised manuscript by updating Section IV and Fig. 2.

Concern 3: HIL personalization requires stronger validation. The human-in-the-loop personalization module is an interesting and potentially important contribution, but the current validation is not fully convincing. The manuscript should explain more clearly how user-specific feedback was used for fine-tuning and how the personalized policy was evaluated

independently.

Response: We thank the reviewer for this comment. During the user study, we collect data from each user during and after immersion. During immersion, we collect self-reported cognitive states feedback ratings, namely Fast Motion Sickness (FMS), cognitive physical load, cognitive mental load, and working memory, together with multimodal objective data (e.g., head-tracking, eye-tracking, and motion), the currently active visual technique, locomotion signals such as walking speed and angular velocity, visual-motion cues, and task-context signals. After immersion, we also collect feedback via a post-task questionnaire that captures technique-specific comfort, perceived visual disruption, preferred intensity, and an overall technique ranking, as described in Section VI. After collecting all the data, before fine-tuning AURORA-HIL, we first preprocess the multimodal objective data by removing outliers beyond three standard deviations using z-score analysis [17], denoising the time series with exponential smoothing [18], and normalizing all features to ensure consistency across sensor modalities. The processed multimodal objective data, user feedback, and task-context signals are then used to construct a user-specific preference profile that adjusts the reward weights in Equation 2 to reflect the cognitive states and preferences of each user. For instance, a participant who is more sensitive to cybersickness but less affected by cognitive load receives a higher weight on cybersickness in the reward. The AURORA policy is then fine-tuned offline with PPO using this user-specific reward, producing the personalized AURORA-HIL policy. We evaluate AURORA-HIL in a separate session, ensuring that the personalized policy is assessed on new, unseen user data, and quantify the benefit of personalization through the direct AURORA versus AURORA-HIL comparison on the same participants, as reported in Table VI.

In the revised manuscript, we have added details on how user-specific feedback and multimodal data are used to construct the preference profile and fine-tune the AURORA-HIL policy in Section III (E), and added the data preprocessing details, including z-score-based outlier removal, exponential-smoothing-based denoising, and feature normalization, in Section IV.

Concern 4: Generalizability beyond the VR maze task is limited. The current evaluation is mainly based on a VR maze navigation task involving walking, coin collection, and digit-span recall. While this is a reasonable experimental paradigm, it is still a relatively narrow scenario. The title and several statements suggest a general framework for adaptive cognitive-state optimization in VR. However, the current evidence supports a more limited

conclusion: AURORA is effective in a walking-based VR maze task with concurrent cognitive demands. The authors should either provide additional evidence across different VR scenarios or narrow the scope of the title, abstract, and conclusion.

Response: We thank the reviewer for this comment. In our proposed framework, the MTL-enabled DL-based prediction model for cognitive-state prediction is trained using the open-source VRWalking dataset [19], which is widely used in SOTA cognitive-state prediction studies [20–22]. To the best of our knowledge, VRWalking is the only publicly available dataset that provides ground-truth labels for all four targeted cognitive states, namely cybersickness, cognitive physical load, cognitive mental load, and working memory, whereas other available datasets provide labels for only a single cognitive state, mostly cybersickness [1, 23, 24]. For this reason, we evaluate our proposed framework on the VRWalking dataset. Since VRWalking was collected using a VR maze simulation, we developed a similar custom VR maze simulation to remain consistent with the environment in which the dataset was collected and to train the RL policy in a setting that aligns with the MTL-enabled DL-based prediction model used for reward shaping. This design maintains consistency between the prediction model and the control policy, ensuring that the predicted cognitive states, reward shaping, and visual technique selection are grounded in the same task context, and it reduces the risk of a mismatch between the prediction and control environments. Beyond the maze-navigation environment, AURORA can be extended to a wide range of immersive VR applications with different levels of visual intensity, interaction styles, and motion patterns, such as roller-coaster simulations, rhythm-based games, shooter games, racing scenarios, exploration environments, waypoint navigation, and sea-voyage experiences [2, 4, 23, 25, 26]. For example, in high-motion VR applications such as roller coasters, shooters, and racing, the MTL-enabled DL-based prediction model can predict cognitive states during immersion, and the RL policy can adaptively select visual techniques (e.g., DGB, DFOV) with appropriate intensity. Similarly, in slower-paced or exploration-focused applications such as beach-city exploration, waypoint navigation, or sea-voyage simulations, the same framework can adjust the visual technique and intensity based on the predicted cognitive states and task context. In each case, the multi-objective formulation jointly balances cybersickness, cognitive load, and working memory rather than optimizing any single state, while remaining sensitive to each application’s interaction and motion characteristics. In the revised manuscript, we have clarified the rationale for the VR maze-navigation evaluation in Section VI and added a discussion on the generalizability of AURORA in Section X. Furthermore, validation across broader VR

tasks and unseen scenarios is already explicitly stated as an important future direction in our limitations (Section VII).

Concern 5: Cognitive-load measurement requires stronger justification. The authors claim that AURORA balances cognitive physical load and cognitive mental load, but these two outcomes appear to be assessed using a single adapted NASA-TLX question administered once per minute during VR navigation. While such a simplified real-time rating may be practically motivated, it cannot provide sufficiently validated evidence for the strong claim that the system balances cognitive load. Moreover, repeated in-task questioning may itself affect cognitive load, working memory, and cybersickness. The authors should justify this measurement choice, clearly distinguish it from the standard NASA-TLX, discuss its limitations, and provide additional behavioral or objective indicators to support the cognitive-load findings.

Response: We thank the reviewer for this comment.

Regarding the justification of cognitive-load measurement: We adopt the same single-item rating protocol used to collect the ground-truth cognitive physical load and cognitive mental load labels in the open-source VRWalking dataset [19] during immersion, which is a widely used protocol for assessing cognitive load during immersion in SOTA cognitive-load measurement studies [21, 22]. Since our MTL-enabled DL-based prediction model is trained on the VRWalking labels, collecting our cognitive-load ratings with the same protocol keeps them consistent with what the model was trained to predict, avoiding any mismatch between the measured cognitive load and the labels used during training. Furthermore, to minimize participant burden, cognitive physical load and cognitive mental load were collected through a single combined prompt rather than two separate prompts. Specifically, we asked participants one question once per minute, *“How physically and mentally demanding is the task on a scale of 1 (not demanding) to 10 (very demanding)? Please provide two ratings, one for each,”* as explicitly stated in Section VI. This allowed us to capture participants’ cognitive physical load and cognitive mental load ratings during VR immersion. Note that, although cognitive physical load and cognitive mental load were presented within a single prompt, participants provided two separate ratings. We have revised Section VI to further clarify that cognitive physical load and cognitive mental load were recorded as separate ratings from the combined prompt.

Regarding NASA-TLX, limitations, and additional behavioral or objective indicators: Prior

work typically used the full six-dimensional NASA-TLX as a pre- and post-task questionnaire, which captures cognitive load only before and after the task and cannot measure how cognitive load evolves during immersion. To address this, recent SOTA cognitive-load measurement studies [21, 22] assess cognitive load during immersion using only the physical-demand and mental-demand dimensions. Therefore, following the same widely used protocol, we measure cognitive physical load and cognitive mental load during immersion. Furthermore, using the full six-dimensional NASA-TLX repeatedly during the task would itself impose substantial cognitive demand. Moreover, our goal is to measure only cognitive load, not the other NASA-TLX dimensions such as frustration or temporal demand, and prior work similarly relies on physical and mental demand for cognitive-load assessment during immersion [21, 22]. For the additional objective indicators, we collected the multimodal objective data (e.g., eye-tracking and head-tracking) and motion data, together with their corresponding subjective labels, such as cybersickness, to train our RL agent for AURORA-HIL. Moreover, as described in Section VI, participants verbally reported their cybersickness on an FMS scale at regular intervals during the simulation, and these subjective ratings are automatically locked with the concurrently recorded objective signals, which serve as objective indicators of cognitive demand.

Concern 6: Statistical analysis and reporting require clarification and revision. The statistical analysis and reporting require further clarification and revision. The use of Shapiro-Wilk tests, Friedman tests, and Wilcoxon signed-rank post hoc comparisons with Holm-Bonferroni correction for cybersickness, cognitive physical load, cognitive mental load, and working memory is generally appropriate for repeated-measures nonparametric data. However, several important issues remain. First, the IPQ presence analysis appears to use Mann-Whitney U tests for pairwise comparisons, although the study seems to involve the same participants across multiple conditions. If this is the case, paired or repeated-measures nonparametric tests, such as Friedman tests followed by Wilcoxon signed-rank tests, would be more appropriate. Alternatively, mixed-effects models could be used. Second, the current analyses do not appear to account for potentially important covariates or confounds, including prior VR experience, baseline sickness level, gender, session/day effects, task familiarization, and condition order. These factors are particularly important because AURORA-HIL was evaluated on a separate day after data from earlier conditions had been used for personalization. Third, the authors should clarify the scope of Holm-Bonferroni correction. It should be clearly stated whether the reported p-values are raw or

adjusted, and whether correction was applied within each outcome or across all pairwise comparisons and outcomes. Fourth, the direction of the reported effect sizes should be defined. For example, the manuscript reports negative effect sizes for some improvements in working memory. The authors should clarify whether the sign reflects the direction of the Wilcoxon statistic or the direction of the experimental effect. Finally, the manuscript should check and correct internal inconsistencies in result reporting. For example, the text states that AURORA achieves the lowest cybersickness score, although AURORA-HIL has a lower mean cybersickness score. Such inconsistencies should be corrected to improve the rigor and interpretability of the findings.

Response: We thank the reviewer for this comment.

Regarding the IPQ analysis: As the IPQ scores were collected from the same participants across conditions, we have followed the reviewer’s suggestion and replaced the Mann–Whitney U test with a Friedman test followed by Wilcoxon signed-rank post hoc comparisons. We adopt this nonparametric repeated-measures approach rather than a mixed-effects model because the ordinal Likert presence scores do not meet the normality assumptions of standard linear mixed-effects models, and because it keeps the presence analysis consistent with the analysis used for our other outcomes. The Friedman test reveals a significant difference in presence across conditions ($\chi^2(6) = 162.36, p < .001, W = 0.576$). Post hoc comparisons show that AURORA achieves significantly higher presence than the baseline ($Z = 5.97, p < .001, r = 0.87$) and all four static techniques (all $p < .001, r = 0.81\text{--}0.86$), and that AURORA-HIL further improves presence over AURORA ($Z = 2.42, p = .016, r = 0.35$). These results support **H2**, indicating that the proposed framework provides a higher sense of presence and a better overall UX than conventional methods. We have updated the corresponding results in Section VI accordingly.

Regarding the Holm-Bonferroni correction: The p -values originally reported in Table VI were the Holm-Bonferroni adjusted values, and the correction was applied within each outcome separately, treating the pairwise comparisons within each of cybersickness, cognitive physical load, cognitive mental load, and working memory as a distinct comparison set, following standard practice for controlling the family-wise error rate [27]. To make this scope explicit, in the revised manuscript we now report both the raw and the Holm-Bonferroni adjusted p -values for each comparison in Table VI, and we state in its caption and in Section VI that the correction was applied within each outcome rather than across all outcomes.

Regarding the direction of the effect sizes: The originally reported rank-biserial r carried a

sign that reflected the direction of the Wilcoxon statistic, which condition in the pair received the higher ranks, rather than the direction of the experimental improvement [28]. Because lower CS, CPL, and CML values and higher WM values all correspond to a better outcome, an improvement appeared as a positive r for CS, CPL, and CML but as a negative r for WM, which could be misread as a worse outcome. In the revised manuscript, we now report the standardized effect size $r = |Z|/\sqrt{N}$, which is unsigned, and indicate the direction of each effect through the sign of the reported Z statistic. We have added this convention to the caption of Table VI, clarifying that the direction of every comparison is given by the sign of Z rather than the sign of r .

Regarding internal consistency: AURORA-HIL has a lower mean cybersickness score ($M = 1.12$) than AURORA ($M = 1.52$), so AURORA-HIL achieves the lowest cybersickness. We have corrected this statement and rechecked the manuscript for similar inconsistencies.

Regarding including prior VR experience, baseline sickness level, gender, session/day effects, and task familiarization: Due to time, budget, and resource constraints, the current study primarily focused on evaluating the effectiveness of the proposed framework, and a controlled investigation of these factors would require a substantially larger and more diverse participant pool. Although these factors are important and can meaningfully influence cybersickness, cognitive load, and working memory, and accounting for prior VR experience, baseline sickness susceptibility, gender, age, session and day effects, and task familiarization would strengthen the analysis, we therefore leave their systematic analysis for future work and have explicitly stated this scope as a limitation in Section VII.

Minor Concern 1: The manuscript contains several grammatical and typographical errors, including repeated phrases, malformed words, and inconsistent terminology, for example, “user cognitivstates in page 7”, “while while the remaining user in page 8”, “. This is less than 27% of the per-frame budget, This is less than 27% of the per-frame budget, remaining within acceptable limits for SOTA real-time VR settings in page 9”. A careful language edit is needed.

Response: We thank the reviewer for this comment. We have carefully reviewed the entire manuscript and corrected the listed errors, including the repeated phrases and malformed words on pages 7–9, along with other typographical and terminology inconsistencies in the revised manuscript.

Minor Concern 2: Some abbreviations should be defined consistently at first use.

Response: We thank the reviewer for this comment. We revised the manuscript to improve abbreviation consistency and readability. Specifically, we expanded machine learning (ML), state-of-the-art (SOTA), and multi-layer perceptron (MLP) at their first use in the main text. We also updated the relevant table captions to make abbreviations self-contained, particularly for CS, CL, CPL, CML, WM, CR, and CR-HIL.

Minor Concern 3: The visual technique library should be described more systematically, including parameter ranges, intensity normalization, and implementation details.

Response: We thank the reviewer for this comment. In our proposed framework, the visual technique library specifies each technique by its underlying rendering parameter and operating range, where DFOV is controlled by the aperture (vignette) fraction (range $[0.2, 1.0]$) via Unity’s Tunneling Vignette Controller, DGB by the Gaussian blur strength (range $[0.0, 1.0]$) via a screen-space Gaussian blur shader, DOF by the focal distance and blur radius (range $[0.5, 10.0]$ m) via the URP Depth-of-Field post-process, and RF by the opacity of the rest-frame overlay (range $[0.0, 1.0]$). For each technique, the native parameter range is linearly mapped to the normalized intensity $I_t \in [0, 1]$ used in the action space and the reward (Equation 2), so that $I_t = 0$ corresponds to no visual technique applied and $I_t = 1$ to the maximum permitted setting. This allows the RL agent to control all techniques on a single comparable scale regardless of their differing native units. We have expanded the description of the visual technique library in Section III(D), specifying each technique by its underlying rendering parameter, its operating range, and how that range is normalized into the common intensity control used by the RL agent.

Minor Concern 4: The limitations section should more explicitly discuss sample size, participant age range, prior VR experience, task specificity, possible order effects, simplified cognitive-load measurement, and the absence of long-term exposure testing.

Response: We thank the reviewer for this comment.

Regarding sample size, participant age range, prior VR experience, and task specificity: Our study setup in Section VI reports the sample size (47 participants), age range (18–39), gender, ethnicity, and prior VR experience of the participants. The limitations in Section VII discuss the specific participant group, the task specificity, and the planned extension to longer exposure durations and a broader visual technique library. Furthermore, we have revised Section VII to discuss these limitations more clearly and to explicitly note the factors that fall outside the current scope and motivate future work.

Regarding possible order effects: In our experimental conditions, we applied counterbalancing to reduce order effects: the first three conditions (e.g., Baseline, Static visual techniques, and AURORA) were counterbalanced across participants. However, the static visual technique sub-conditions (e.g., DFOV, DGB, DOF, and RF) within Condition 2 were presented in randomized order to minimize potential ordering effects among these four static visual techniques. The only exception was Condition 4, AURORA-HIL, which was necessarily conducted last, on a separate day, because human-in-the-loop RL requires prior interaction data from which to learn user-specific preferences before the policy can be adapted. Since the fine-tuning is performed offline via PPO updates on data collected during the AURORA condition, the personalized policy first had to be trained before it could be evaluated in the following session (AURORA-HIL). We have revised Section VI to remove the sequential phrasing and explicitly clarify the scope of counterbalancing used to reduce order effects.

Regarding the justification of cognitive-load measurement: We adopt the same single-item rating protocol used to collect the ground-truth cognitive physical load and cognitive mental load labels in the open-source VRWalking dataset [19] during immersion, which is a widely used protocol for assessing cognitive load during immersion in SOTA cognitive-load measurement studies [21, 22]. Since our MTL-enabled DL-based prediction model is trained on the VRWalking labels, collecting our cognitive-load ratings with the same protocol keeps them consistent with what the model was trained to predict, avoiding any mismatch between the measured cognitive load and the labels used during training. Furthermore, to minimize participant burden, cognitive physical load and cognitive mental load were collected through a single combined prompt rather than two separate prompts. Specifically, we asked participants one question once per minute, *“How physically and mentally demanding is the task on a scale of 1 (not demanding) to 10 (very demanding)? Please provide two ratings, one for each,”* as explicitly stated in Section VI. This allowed us to capture participants’ cognitive physical load and cognitive mental load ratings during VR immersion. Note that, although cognitive physical load and cognitive mental load were presented within a single prompt, participants provided two separate ratings. We have revised Section VI to further clarify that cognitive physical load and cognitive mental load were recorded as separate ratings from the combined prompt.

Minor Concern 5: The manuscript should avoid overclaiming that AURORA generally optimizes cognitive state in VR unless additional task scenarios are evaluated.

Response: We thank the reviewer for this comment. In our proposed framework, the MTL-

enabled DL-based prediction model for cognitive-state prediction is trained using the open-source VRWalking dataset [19], which is widely used in SOTA cognitive-state prediction studies [20–22]. To the best of our knowledge, VRWalking is the only publicly available dataset that provides ground-truth labels for all four targeted cognitive states, namely cybersickness, cognitive physical load, cognitive mental load, and working memory, whereas other available datasets provide labels for only a single cognitive state, most commonly cybersickness [1, 23, 24]. For this reason, we evaluate our proposed framework on the VRWalking dataset. Since VRWalking was collected using a VR maze simulation, we developed a similar custom VR maze simulation to remain consistent with the environment in which the dataset was collected and to train the RL policy in a setting that aligns with the MTL-enabled DL-based prediction model used for reward shaping. This design maintains consistency between the prediction model and the control policy, ensuring that the predicted cognitive states, reward shaping, and visual technique selection are grounded in the same task context, and it reduces the risk of a mismatch between the prediction and control environments. Beyond the maze-navigation environment, AURORA can be extended to a wide range of immersive VR applications with different levels of visual intensity, interaction styles, and motion patterns, such as roller-coaster simulations, rhythm-based games, shooter games, racing scenarios, exploration environments, waypoint navigation, and sea-voyage experiences [2, 4, 23, 25, 26]. For example, in high-motion VR applications such as roller coasters, shooters, and racing, the MTL-enabled DL-based prediction model can predict cognitive states during immersion, and the RL policy can adaptively select visual techniques (e.g., DGB, DFOV) with appropriate intensity. Similarly, in slower-paced or exploration-focused applications such as beach-city exploration, waypoint navigation, or sea-voyage simulations, the same framework can adjust the visual technique and intensity based on the predicted cognitive states and task context. In each case, the multi-objective formulation jointly balances cybersickness, cognitive load, and working memory rather than optimizing any single state, while remaining sensitive to each application’s interaction and motion characteristics. In the revised manuscript, we have clarified the rationale for the VR maze-navigation evaluation in Section VI and added a discussion on the generalizability of AURORA in Section X. We also note that validation across broader VR tasks and unseen scenarios is already identified as an important future direction in the limitations (Section VII).

II. RESPONSES TO THE COMMENTS FROM REVIEWER-2

Concern 1: Conceptual Rationale for Cognitive Load, CML , and CPL , and Link to User Experience

Concern 1.1: Lack of justification for splitting cognitive load. The manuscript introduces Cognitive Mental Load (CML) and Cognitive Physical Load (CPL) as separate metrics derived from the general cognitive load discussed in the Introduction and Background. Although the Introduction lists these metrics as part of the contribution, the paper does not explain why cognitive load should be divided into these two components. From a psychological standpoint, cognitive load is a single construct with intrinsic, extraneous, and germane components (Sweller, 1988) and cannot be meaningfully separated into “mental” and “physical” load.

Concern 1.2: Insufficient operationalization of UX and its relationship with the selected cognitive-state metrics. The manuscript repeatedly claims that AURORA improves the overall user immersive experience (UX), but UX is not clearly defined or operationalized. While cybersickness is evidently relevant to comfort and immersion, the paper does not explain how the three selected targets—cybersickness, cognitive load, and working memory—jointly constitute or predict UX. More specifically, these metrics represent different types of constructs: cybersickness is a discomfort/symptom measure, cognitive load is a workload-related construct, and working memory is closer to task performance or cognitive capacity. The manuscript treats them as a unified optimization target, but does not provide a clear human-factors or HCI rationale for how improvements in these measures should be interpreted as improvements in UX. This issue is especially important because the RL framework involves trade-offs among these metrics. For example, a technique may reduce cybersickness while increasing workload or impairing task performance. Without defining UX and explaining how these trade-offs are resolved from a user-experience perspective, the claim that the framework optimizes “overall UX” remains under-specified.

Recommendation: The authors should substantially revise the conceptual framing of cognitive load and UX. First, they should justify why cognitive load is decomposed into CML and CPL, and clarify whether CPL is intended to represent cognitive load, physical demand, or a broader workload component. If CPL is derived from the NASA-TLX Physical Demand dimension, it should not be labeled as “cognitive” without theoretical justification. Second,

the authors should define UX explicitly and explain how cybersickness, cognitive load, and working memory jointly contribute to this outcome. The revised manuscript should ground these constructs in established cognitive load theory and HCI/user-experience literature, and clearly distinguish between cognitive states, workload dimensions, task performance, and user-experience outcomes.

Response: We thank the reviewer for this comment.

Regarding Concern 1.1: To measure cognitive load, prior work typically relied on pre- and post-task questionnaires (e.g., the six-dimensional NASA-TLX or the single 9-point Paas scale), which capture cognitive load only before and after immersion and cannot measure how it evolves during immersion [29–31]. To address this, recent SOTA cognitive-load measurement studies assess cognitive load using only the physical-load and mental-load dimensions of the NASA-TLX during immersion [19, 21, 22]. Furthermore, the VRWalking dataset [19], which we use to train our proposed framework AURORA, provides separate ground-truth labels for cognitive physical load and cognitive mental load and is widely used in SOTA cognitive-state prediction studies [20–22]. We therefore measure cognitive physical load and cognitive mental load separately, consistent with the VRWalking dataset, to avoid any mismatch between the measured cognitive load and the labels used during training. Furthermore, separating cognitive load into cognitive physical load and cognitive mental load is important for active VR locomotion, because the two are driven by different factors and may respond differently to applied visual techniques. Specifically, cognitive physical load is associated with compensatory head and body movement, postural control, and locomotion effort, whereas cognitive mental load is associated with navigation, spatial reasoning, and interpreting degraded or incomplete visual information [12, 14]. For example, the DFOV visual technique may increase cognitive physical load by requiring additional head movement, whereas the DGB visual technique may increase cognitive mental load by making visual cues harder to interpret. Combining these two dimensions into a single value would hide such differing effects and remove the signal the RL agent needs to balance visual comfort, movement effort, and task-related cognitive processing. Therefore, in this study we separately measure cognitive physical load and cognitive mental load, which provides a more informative representation for adaptive visual-technique selection in VR.

Regarding Concern 1.2: In our work, we define User Immersive Experience (UIX) as the overall quality of a user’s experience in an immersive VR environment, reflecting how comfortable,

mentally and physically manageable, and engaging the experience feels during immersion, which is widely studied in SOTA VR immersive-experience and HCI studies [1, 2, 32]. Following these works, we built a custom 5-point Likert questionnaire of six statements to assess how cybersickness, cognitive load, and working memory jointly contribute to the overall UX, as well as the quality of the adaptive selection, the resulting overall experience, and the user’s willingness to adopt the framework. Since we are not interested in isolating the individual influence of each cognitive state on UX, we designed these statements to capture how the jointly optimized cognitive states shape the overall UX. For instance, **(S1)** “*AURORA better balanced my discomfort, cognitive demand, and working memory than the conventional methods*” measures whether AURORA jointly reduces cybersickness, balances cognitive load, and improves working memory to achieve a better UX. Similarly, **(S5)** “*Using AURORA improved my overall VR experience compared to sessions with conventional methods*” measures the resulting overall UX, and **(S6)** “*I would recommend AURORA to others because of its adaptive approach compared to conventional methods*” measures the user’s willingness to adopt the framework. Furthermore, from the user study, we found that approximately 89% of participants agreed or strongly agreed across all six statements, indicating that jointly balancing the four cognitive states yields a better overall UX than conventional methods. In addition, we also measured the sense of presence through the 7-point Likert IPQ questionnaire after each condition [33], following widely used presence measurement studies [1, 2, 32]. Since presence is closely related to UX and is a widely used indicator of immersive-experience quality, the higher presence scores achieved by AURORA further confirm the improvement in overall UX, as explicitly stated in Section VI of the manuscript.

Concern 2: Ambiguity in RL State Feature Encoding. The RL state vector includes multi-modal sensor features and Transformer-predicted cognitive states. However: The manuscript does not provide a high-level explanation of how raw sensor data are processed, normalized, or combined into the RL state vector. Each feature appears arbitrarily mapped, with no methodological justification in the text. Although the system is open-source and reproducible from code, the manuscript alone does not allow readers to interpret how RL actions relate to cognitive states and rewards. **Recommendation:** Include a high-level diagram or description illustrating how multi-modal features and Transformer outputs form the RL state.

Response: We thank the reviewer for this comment. In our proposed framework, the RL state

is built from the streaming multimodal data (e.g., eye-tracking, head-tracking, etc.), along with locomotion data (e.g., walking speed and angular velocity), a visual motion cue (optic flow magnitude), and a task-context indicator of whether the user is currently turning or walking straight, together with the last applied visual technique and its intensity. After collecting all the data, before training AURORA, we preprocess the collected data to ensure consistency between the inputs to the MTL-enabled DL-based prediction model and the RL agent. Specifically, the multimodal data and the continuous derived data (locomotion and optic-flow features) are preprocessed by removing outliers beyond three standard deviations using z-score analysis [17], denoising the time series with exponential smoothing [18], and normalizing all features to ensure consistency across modalities. The categorical task-context indicator and the discrete applied technique and intensity are then encoded directly, before all components are combined into a single fixed-length state vector. It is important to note that the cognitive states predicted by the MTL-enabled DL-based prediction model, namely cybersickness, cognitive physical load, cognitive mental load, and working memory, are not included in the RL state. Instead, these predicted states are normalized to the range $[0, 1]$ using Equation 1 and are used only in the reward computation in Equation 2. The RL state therefore represents the user’s current behavior and interaction context, whereas the predicted cognitive states represent the response to the applied visual technique. This separation prevents the reward target from being embedded directly into the state and preserves a clear distinction between what the agent observes and how its actions are evaluated. The selected feature groups are likewise motivated by their relevance to the targeted cognitive states, so that each group informs at least one state that the RL agent must balance. Specifically, head motion and optic flow are well-established contributors to cybersickness [13, 14, 34], whereas pupil diameter, gaze direction, and fixation stability reflect cognitive mental load [14, 35]. Similarly, locomotion and task context capture cognitive physical load and the motion characteristics associated with active walking and turning [12, 14]. In addition, working memory is affected by the same perceptual and cognitive resources involved in cybersickness and cognitive load [14, 36, 37]. In the revised manuscript, we have expanded the state definition in Section III(C) to describe how the multimodal features are combined into the RL state vector, and added the data preprocessing details, including z-score-based outlier removal, exponential-smoothing-based denoising, and feature normalization, in Section IV. We have also updated the framework overview (Fig. 1) to better illustrate how the multimodal data form the RL state and how the MTL-enabled DL-based prediction model shapes the reward for

the RL agent.

Concern 3: Clarification Needed on the User Study Design and Repeated-Measures Structure. The user study appears to be a within-subjects repeated-measures design, since the manuscript states that participants completed all study conditions, including the baseline, static techniques, AURORA, and AURORA-HIL. The use of Friedman tests and Wilcoxon signed-rank tests also implies a paired/repeated-measures structure. However, the current description is somewhat unclear. The paper first describes four broad experimental conditions: baseline, static visual techniques, AURORA, and AURORA-HIL. Yet in the results, the four static techniques—DFOV, DGB, DOF, and RF—are analyzed separately, resulting in seven analyzed levels: Baseline, DFOV, DGB, DOF, RF, AURORA, and AURORA-HIL. This ambiguity should be clarified because the interpretation of the statistical tests depends on the exact experimental structure. In particular, the authors should specify: whether all 47 participants experienced all seven analyzed conditions; whether DFOV, DGB, DOF, and RF were treated as separate repeated-measures conditions or as subconditions within a single “static techniques” condition; how the order of conditions was randomized or counterbalanced; whether potential order effects, fatigue effects, learning effects, or carryover effects were controlled, especially given that AURORA-HIL was conducted on a separate day with the same participants. This is especially important because VR sickness, task familiarity, and working memory performance can all be sensitive to exposure order and accumulated fatigue. The study design is potentially valid, but the manuscript should describe the repeated-measures structure and counterbalancing procedure more explicitly.

Response: We thank the reviewer for this comment and apologize for the ambiguity.

Regarding the repeated-measures structure: In our study, all 47 participants experienced four conditions, namely the Baseline condition, the Static visual techniques condition, the AURORA condition, and the AURORA-HIL condition, as explicitly stated in the manuscript section VI. Among these, the Static visual techniques condition contains four subconditions (e.g., DFOV, DGB, DOF, and RF). We treat these as a single condition because each applies a fixed, predefined intensity without jointly optimizing the user’s current cognitive state. It is important to note that while each condition lasted 10 minutes, the Static visual techniques condition consisted of four subconditions that each participant performed for 10 minutes each (40 minutes in total), one after another in a randomized order, following prior works [1, 2]. Since each of the four static

subconditions was presented and analyzed separately, together with the Baseline, AURORA, and AURORA-HIL conditions, we report those subconditions separately in Tables IV–VI. To avoid this ambiguity and make the grouping explicit, we have revised Tables IV–VI in the revised manuscript to label the four static subconditions as Static (DFOV), Static (DGB), Static (DOF), and Static (RF).

Regarding the order and counterbalancing: In our experimental conditions, we applied counterbalancing to reduce order effects: the first three conditions (e.g., Baseline, Static visual techniques, and AURORA) were counterbalanced across participants. However, the static visual technique subconditions (e.g., DFOV, DGB, DOF, and RF) within Condition 2 were presented in randomized order to minimize potential ordering effects among these four static visual techniques, following prior works [1, 2]. Between every condition, along with the static subconditions, participants were given a rest and were allowed to rest for as long as needed until they felt ready to continue without discomfort. The only exception was Condition 4, AURORA-HIL, which was necessarily conducted last, on a separate day, because human-in-the-loop RL requires prior interaction data from which to learn user-specific preferences before the policy can be adapted. Since the fine-tuning is performed offline via PPO updates on data collected during the AURORA condition, the personalized policy first had to be trained before it could be evaluated in the following session (AURORA-HIL). It is worth mentioning that prior work has shown that cybersickness symptoms can persist from hours up to several days after VR exposure (<https://doi.org/10.1007/s10055-016-0285-9>, <https://doi.org/10.3357/ASEM.2394.2009>), suggesting that participants may still experience residual cybersickness symptoms the following day. Consistent with this, in our study, a few participants felt discomfort both on the day of Conditions 1–3 and continued to experience it on the following day, yet reported substantially greater comfort during the AURORA-HIL condition. For instance, after AURORA-HIL, one participant noted: *“I was still feeling uncomfortable from the previous day, but during today’s session I felt more comfortable, and after the session I felt better than I had the day before.”*. It is important to note that the sequential wording in the paper (e.g., Initially, Next, Then, and Finally) was intended only to describe the order in which conditions are introduced in the manuscript, not the actual exposure order in the user study. We have revised Section VI to remove the sequential phrasing, explicitly clarify the scope of counterbalancing, explain the rationale for evaluating AURORA-HIL separately after the non-HIL conditions, and added participant quotes to Section VI(B) in

our revised manuscript.

III. RESPONSES TO THE COMMENTS FROM REVIEWER-3

Concern 1: The paper claims to be the “first” work to model the optimization of a VR user’s cognitive state as a multi-objective RL problem. However, AURORA’s primary contribution lies in jointly predicting multiple cognitive states and utilizing them for reward shaping; yet, this fundamentally constitutes a combination of multi-task prediction and RL, lacking any groundbreaking theoretical innovation.

Response: We thank the reviewer for this comment. In our proposed framework, the core contribution is a closed-loop VR framework that jointly predicts and optimizes multiple user cognitive states during immersion by integrating an MTL-enabled DL-based prediction model, a PPO-based RL agent, an adaptive visual technique engine, and HIL-based personalization. Specifically, the MTL-enabled DL model predicts cybersickness, cognitive physical load, cognitive mental load, and working memory from VR streaming data. These predictions are then used to construct a multi-objective reward that guides the RL agent to adaptively select visual techniques from our developed visual technique library and adjust their intensities. This enables AURORA to move beyond single-state prediction and fixed-intensity mitigation by learning context-aware policies that reduce cybersickness, balance cognitive load, and improve working memory during immersion. Prior work on RL in consumer electronics demonstrates that the contribution of applied RL systems often lies in formulating a domain-specific decision problem and integrating RL effectively into the target system, rather than proposing a new groundbreaking learning algorithm [38–41]. For instance, Tiwari et al. [38] used federated RL for consumer-centric cyborg-IoMT applications, where RL enables adaptive decision-making in dynamic mobile and fog environments. Following this applied RL-based optimization paradigm, AURORA uses RL not as a standalone theoretical contribution, but as the adaptation mechanism for a distinct and unexplored VR problem that jointly optimizes users’ cognitive states during immersion through adaptive visual-technique selection and continuous intensity control. Specifically in VR, users perform concurrent tasks during immersion, including physical movement, complex visual-scene processing, and task-level cognitive reasoning. These simultaneous demands can increase cybersickness, elevate cognitive physical and mental load, and reduce working memory. Moreover, these cognitive states are interconnected through shared perceptual and cognitive resources, so im-

proving one state can come at the expense of another. Therefore, jointly optimizing them requires more than predicting a single state or applying a fixed mitigation technique. Prior ML and DL models for VR user-state estimation, however, are typically trained for a single state, most often cybersickness, which limits their ability to capture the shared dependencies among cybersickness, cognitive load, and working memory [7, 19, 21, 23, 42–48]. More importantly, these works are primarily predictive and do not use the predicted states to drive adaptive visual-technique selection at an appropriate intensity. More recent works, such as [1, 2], predict cybersickness severity and map explanations to predefined visual techniques, but they remain single-objective, rely on fixed-intensity presets, and do not optimize multiple cognitive states simultaneously. As a result, a visual technique that reduces cybersickness may still increase cognitive load or impair working memory, thereby degrading the overall UX. For example, narrowing the field of view can reduce cybersickness, but it may also increase cognitive load and impair working memory by limiting visual context and increasing task difficulty [12–14]. Addressing this limitation requires an adaptive approach that not only triggers a visual technique automatically, but also adjusts its intensity based on the user context and the individual user’s response during immersion. RL is well-suited to this VR-domain problem because it supports sequential decision-making and optimization through interaction-driven feedback, and it has been applied across diverse VR domains such as education, training, and healthcare to deliver adaptive, user-centered experiences [49–51]. Building on these observations, and to the best of our knowledge for the first time, we use RL in the VR domain to jointly optimize multiple user cognitive states by adaptively selecting visual techniques and controlling their continuous intensity. We formalize this problem as a multi-objective Markov Decision Process (MDP) represented by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, in which the state $\mathcal{S}$ captures VR observations and context informative of the targeted cognitive states, the action $\mathcal{A}$ jointly selects a visual technique and its continuous intensity, and the reward R , derived from the MTL-enabled DL-based prediction model, balances cybersickness, cognitive physical load, cognitive mental load, and working memory through Equations 1 and 2. Thus, AURORA is not merely a combination of multi-task prediction and RL. Multi-task prediction alone estimates user states but does not determine how to adapt the VR environment, whereas RL alone does not define which competing cognitive states should be optimized, how they should be balanced, or how visual techniques and intensities should be controlled during immersion. Our contribution is the closed-loop integration of these components into a VR-domain-specific multi-objective optimization framework that jointly predicts and optimizes competing cognitive

states through adaptive technique-and-intensity control, further personalized through HIL-based fine-tuning. AURORA is therefore, to the best of our knowledge, the first framework to formulate and implement the joint optimization of competing VR user cognitive states as a multi-objective RL problem during immersion.

Concern 2: The multi-task DL model was trained on the VRWalking dataset, while the RL agent was trained in a domain-randomized simulation; however, the RL reward function directly relies on the predictive outputs of this DL model. Nevertheless, the paper neither evaluates how the prediction errors of this DL model are amplified or propagated during RL training, nor provides an analysis of the RL agent’s online prediction errors within a real-world user environment.

We thank the reviewer for this comment. In our proposed framework, the MTL-enabled DL-based prediction model is trained on the VRWalking dataset [19], which was collected in a VR maze environment. Since this model is used for RL reward shaping, we train the RL agent in a similar VR maze simulation consistent with the same environment, as explicitly stated in Section IV. Although the domain randomization varies conditions such as layout and walking speed, these variations remain within the same maze setting, so the MTL-enabled DL-based prediction model still operates on the kind of data it was trained on during RL training. This consistency limits the propagation of prediction errors into the reward. Furthermore, to avoid optimizing an unreliable proxy from the MTL-enabled DL-based prediction model, we first validated the model across all four cognitive states using 10-fold cross-validation, where it achieves 93%, 88%, 94%, and 90% accuracy for cybersickness, cognitive physical load, cognitive mental load, and working memory, respectively, with strong per-class F1 scores as shown in Table III. This strong performance suggests that the reward signal is based on a reliable predictor rather than a highly noisy or poorly calibrated one. In addition, as shown in Fig. 2, the reward for RL agent rises steadily and converges to a stable plateau during training, rather than oscillating or diverging, suggesting that prediction errors are not being amplified during RL training. Moreover, the learned policy was evaluated independently in the user study, where the four cognitive states are measured directly from participant feedback ratings during immersion rather than through the model. As reported in Tables IV and V, AURORA reduced cybersickness, balanced cognitive load, and improved working memory relative to the baseline. These results indicate that, in our experimental setting, the MTL-enabled DL-based reward signal was sufficiently aligned with the participants’ actual cognitive states, with no substantial prediction error propagating into the

learned policy.

Concern 3: The description of the HIL personalization process is vague and lacks reproducibility. Furthermore, critical details, such as the offline fine-tuning procedure, data volume, and number of fine-tuning rounds, are missing, rendering the method impossible to reproduce or to evaluate for its effectiveness.

Response: We thank the reviewer for this comment. In our proposed framework, the HIL personalization is performed separately for each participant using the data collected during their own 10-minute Condition 3 (AURORA) session, since the first two conditions (Baseline and static techniques) do not involve the RL agent. During this session, we collect the participant’s self-reported cognitive-state ratings, namely Fast Motion Sickness, cognitive physical load, cognitive mental load, and working memory, together with the multimodal data (e.g., head-tracking, eye-tracking, and motion), the active visual technique, locomotion signals such as walking speed and angular velocity, visual-motion cues, and task-context signals. After the session, the participant also completes a post-task questionnaire that captures technique-specific comfort, perceived visual disruption, preferred intensity, and an overall technique ranking, as described in Section VI. After collecting all the data, before fine-tuning AURORA-HIL, we preprocess the multimodal data by removing outliers beyond three standard deviations using z-score analysis [17], denoising the time series with exponential smoothing [18], and normalizing all features to ensure consistency across modalities. The predictions from the MTL enable DL model, together with the reported cognitive states, the task-context signals, and the post-task questionnaire, are then used to construct a user-specific preference vector that personalizes the reward in Equation 2 for that participant, adjusting the reward weights, the technique-specific intensity bounds, and the relative preferences over visual techniques. For instance, a participant who is more sensitive to cybersickness or finds high-intensity DGB uncomfortable receives a higher weight on cybersickness and a reduced allowable DGB intensity, while the policy continues to optimize across all four cognitive states. The fine-tuning is performed offline with PPO under this personalized reward. Rather than training from scratch, we start from the already-trained AURORA policy and reuse the PPO hyperparameters reported in Table II, running for a reduced budget of approximately 5×10^4 update steps, compared with the 5×10^5 steps used for full training, over a single fine-tuning round to produce the personalized AURORA-HIL policy. We then evaluate AURORA-HIL in a separate session on new, unseen data from the same participant, so that the personalized policy is never evaluated on the data used to personalize it,

and we quantify the benefit of personalization through the direct AURORA versus AURORA-HIL comparison on the same participants, as reported in Table VI. In the revised manuscript, we have added these details on how the user-specific preference vector is constructed and how the AURORA-HIL policy is fine-tuned in Section III(E). We also updated Table II to report the fine-tuning budget, and added the data preprocessing details in Section IV.

Concern 4: The sequence was fixed as Baseline $\rightarrow$ Static $\rightarrow$ AURORA $\rightarrow$ AURORA-HIL. No Latin square or randomization procedures were employed, resulting in significant order and fatigue effects. Furthermore, participants were aware of the sequence of conditions, which may have introduced expectation bias.

Response: We thank the reviewer for this comment and apologize for the confusion.

In our experimental conditions, we applied counterbalancing to reduce order effects. Specifically, the first three conditions (e.g., Baseline, Static visual techniques, and AURORA) were counterbalanced across participants. However, the static visual technique subconditions (e.g., DFOV, DGB, DOF, and RF) within Condition 2 were presented in randomized order to minimize potential ordering effects among these four static visual techniques, following prior works [1, 2]. The only exception was Condition 4, AURORA-HIL, which was necessarily conducted last, on a separate day, because human-in-the-loop RL requires prior interaction data from which to learn user-specific preferences before the policy can be adapted. Since the fine-tuning is performed offline via PPO updates on data collected during the AURORA condition, the personalized policy first had to be trained before it could be evaluated in the following session (AURORA-HIL). It is important to note that the sequential wording in the paper (e.g., Initially, Next, Then, and Finally) created the impression of a fixed exposure sequence (Baseline $\rightarrow$ Static $\rightarrow$ AURORA $\rightarrow$ AURORA-HIL), whereas it was intended only to describe the order in which the conditions are introduced in the manuscript, not the actual exposure order in the user study. We have revised Section VI to remove the sequential phrasing that suggested a fixed exposure order and to explicitly clarify the scope of counterbalancing.

Concern 5: Table VI employs the Wilcoxon signed-rank test; however, this test requires that paired data originate from the same distribution and exhibit symmetry—a condition that was not verified. Furthermore, the coefficient r is interpreted as the “rank-biserial correlation,” yet the table fails to provide the corresponding Z-values. Moreover, some r values approach 1 or exceed it, which is theoretically impossible outcomes, suggesting errors in either the calculation or the reporting of the results.

Response: We thank the reviewer for these detailed comments on Table VI.

Regarding the coefficient r : In the manuscript Table VI, r denoted the matched-pairs rank-biserial correlation, computed as $r = (W^+ - W^-)/(W^+ + W^-)$, where W^+ and W^- are the sums of positive and negative signed ranks, following prior work [28, 52]. This coefficient is bounded on $[-1, 1]$ by construction and reaches exactly ± 1.00 when all non-zero paired differences share the same sign. This is precisely the case for the DFOV vs. DGB comparison on CPL, where all 47 participants had higher CPL under DFOV, so $W^- = 0$ and $r = 1.00$, which is the defined boundary of the coefficient under complete directional dominance rather than a calculation or reporting error. Because this coefficient is computed directly from W^+ and W^- and already carries a sign indicating direction, the corresponding Z -statistic was not required to obtain or interpret it. However, this convention shares the same symbol r with the standardized Wilcoxon effect size $r = |Z|/\sqrt{N}$, which is computed from Z and is the more commonly expected convention when Z is reported alongside a Wilcoxon test. Following the reviewer’s suggestion, we have revised Table VI to report the standardized effect size r instead of the matched-pairs rank-biserial correlation. Since this measure does not itself convey direction, Z is now necessary both to compute the effect size and to indicate direction through its sign. We have therefore replaced r with the standardized Wilcoxon effect size and added Z and raw p -values to Table VI in the revised manuscript.

Regarding symmetry: we verified the symmetry of the paired-difference distributions for each comparison using the Miao–Gel–Gastwirth symmetry test [53], and found no significant evidence of asymmetry in the paired differences (all $p > .05$), supporting the use of the Wilcoxon signed-rank test for the post hoc pairwise comparisons. We report these symmetry test results in Section VI of the revised manuscript.

Concern 6: The comparative methods are limited to static techniques and a non-adaptive baseline, lacking comparisons against other adaptive approaches. Consequently, it cannot be demonstrated that AURORA’s multi-objective RL strategy significantly outperforms simpler adaptive strategies.

Response: We thank the reviewer for this comment. Prior cybersickness mitigation approaches can be broadly categorized as static or rule-based adaptive methods. In static approaches, a single

mitigation technique is applied at a fixed, predefined intensity regardless of the user’s current cognitive state [3–8]. In contrast, rule-based adaptive methods adjust the mitigation based on a specific predicted signal, such as cybersickness severity or optic flow, using predefined thresholds or handcrafted mappings [1, 2, 9–11]. For instance, explainable AI (XAI)-guided rule-based methods [1, 2] use model explanations to identify important features and trigger predefined mitigation actions. However, these methods remain limited because the mitigation policy is manually designed, where the selected technique, activation threshold, and intensity are fixed in advance and cannot adapt continuously to the user’s evolving cognitive and physiological state. Moreover, existing static and rule-based approaches primarily target cybersickness alone and do not jointly account for other cognitive states, such as cognitive load and working memory [1–10]. As a result, they cannot balance multiple competing objectives, such as reducing cybersickness while balancing cognitive load and improving working memory. In some cases, a cybersickness-targeted mitigation response may reduce cybersickness but increase cognitive load or interfere with working memory, thereby degrading the overall experience. For this reason, we do not consider these rule-based methods as fully adaptive in the sense of our work, since they lack a learned policy that continuously balances the competing cognitive states during immersion, as clarified in the Abstract and Section I. Accordingly, to evaluate our proposed framework, we compare AURORA against static visual techniques in Section VI as the fixed-intensity baseline, where each technique is applied at a predefined intensity without regard to the user’s real-time predicted cognitive state. While a direct comparison against rule-based dynamic mitigation approaches is not included in the current user study, we acknowledge this as a limitation and identify it as a direction for future work in Section VII.

Concern 7: RL training is conducted within a domain-randomized simulation; however, the user state transition model within this simulation is unknown and has not been fitted using real user data. Consequently, whether the resulting policy can effectively operate on real users remains, in essence, unverified.

Response: We thank the reviewer for this comment. Our proposed framework is trained using the open-source VRWalking dataset [19], which is widely used in SOTA cognitive-state prediction studies [20–22]. To the best of our knowledge, VRWalking is the only publicly available dataset that provides ground-truth labels for all four targeted cognitive states, namely cybersickness, cognitive physical load, cognitive mental load, and working memory, whereas other available datasets provide labels for only a single cognitive state, most commonly cybersickness [1, 23, 24].

This dataset was collected from 39 real users navigating a virtual maze environment, and it provides eye-tracking and head-tracking data together with ground-truth labels for our four target cognitive states. To remain consistent with the environment in which VRWalking was collected, we developed a custom VR maze simulation so that the RL policy is trained in a setting that aligns with the MTL-enabled DL-based prediction model used for RL reward shaping. This design keeps the MTL-enabled DL-based prediction model and the RL policy grounded in the same task context, reducing the risk of a mismatch between the environment used for prediction and the one used for control. Within this maze environment, we apply domain randomization that varies conditions such as layout, motion dynamics, visual flow, and walking speed, as described in Section III(C), to improve generalization across users and environments [54] rather than committing to a single fixed assumption about how users respond to visual techniques and intensity changes. Even though the training environment is domain-randomized, both the RL agent and the reward remain grounded in real user data throughout training, since the eye-tracking and head-tracking data underlying the simulation, as well as the cognitive-state prediction used in the reward, come from VRWalking. Domain randomization is applied only to aspects that VRWalking cannot capture, such as how the environment would respond to a new visual technique or intensity chosen by the agent, while the real data continues to inform the RL agent’s observations and the reward throughout training. Importantly, we directly verify that the learned policy operates effectively on real users. Beyond the technical evaluation in Section V, the policy is validated in our user study (Section VI) with 47 participants on a consumer headset, where the targeted cognitive states are measured directly from participant feedback ratings during immersion rather than from the model. In this study, AURORA reduced cybersickness while lowering cognitive load and improving working memory, confirming that the policy transfers effectively from the domain-randomized training environment to real users. Finally, the policy is further adapted to each user through HIL fine-tuning (Section III(E)). Using each user’s in-session responses and stated preferences, we adjust the reward weights and intensity bounds and fine-tune the deployed policy, so that it is corrected toward that user’s actual responses rather than remaining fixed to the training-time dynamics.

Minor Concern 8: The terms “cognitive mental load” and “CML” are used interchangeably within the main text; however, in Tables I and III, “CML” is inconsistently mixed with “cognitive mental load.” Furthermore, Fig. 2 introduces “ r_{CML} ” and “ r_{CPL} ,” yet these terms are not explicitly defined in the caption.

Response: We thank the reviewer for this comment. We used “CL” in Table I to denote the general cognitive-load construct for the high-level comparison of prior work, whereas CPL (cognitive physical load) and CML (cognitive mental load) denote its two measured sub-components, widely used in SOTA cognitive-state measurement studies [19–22]. Furthermore, in Fig. 2, “ r_- ” denotes the reward contribution of each state, such that r_{CPL} and r_{CML} refer to the reward terms associated with CPL and CML. To avoid confusion with the state abbreviations used elsewhere in the paper, we have removed this prefix and updated the figure caption to explicitly define CS, CPL, CML, and WM in the revised manuscript.

Minor Concern 9: The units or normalization method for I_t are not explained in the reward definition.

Response: We thank the reviewer for this comment. In the reward definition (Equation 2), I_t denotes the normalized intensity of the active visual technique at timestep t , defined on $[0, 1]$, where $I_t = 0$ corresponds to no mitigation and $I_t = 1$ to the maximum permitted setting for the selected technique. Each technique’s parameter (e.g., the aperture fraction for DFOV or the Gaussian blur strength σ for DGB) is linearly mapped to this common $[0, 1]$ scale, so that the intensity penalty $\lambda_m I_t$, the abrupt-change penalty $\lambda_\Delta |I_t - I_{t-1}|$, and the stability term $\lambda_{stab}(1 - |I_t - I_{t-1}|)$ are comparable across techniques regardless of their differing native units. In the revised manuscript, we have clarified that I_t in Equation 2 is the normalized technique intensity on $[0, 1]$, and we describe its per-technique normalization and parameter ranges in Section III(D).

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